

Researchers, for instance at Fermilab and Argonne, theorise that by placing quantum processors in these cavities, the qubits will be able to interact undisturbed for seconds rather than the current record of milliseconds, giving them enough time to perform complex calculations.

The photons are encoded using thermo-optical phase shifters that are controlled by electrical voltages. Different settings of the phase shifters control the photon's transmission behaviours in the interferometers, enabling different qubit-state encoding and different quantum operations. To scale up the system to something truly useful, the researchers will have to figure out a way to generate many more identical, entangled photons on the chip.

There is also the engineering challenge of fitting enough phase shifters, beam splitters, and other optical components onto the chip to handle all those photons. However, silicon photonics has shown the capacity for cramming many devices into tight spaces and getting them all to work with high precision and appears to be the practical way to implement the ultimate large-scale photonic quantum processor.

Graphene bolometer. Physicists at Aalto University and VTT Technical Research Centre of Finland have developed a new detector for measuring energy quanta at an unprecedented resolution. This type of detector, called a bolometer, measures the energy of incoming radiation by measuring how much it heats up the detector. Measuring the energy of qubits is at the heart of how quantum computers operate. Most quantum computers currently measure a qubit's energy state by measuring the voltage induced by the qubit.

Bolometers are now entering the field of quantum technology and perhaps their first application could be in reading out the quantum information from qubits. Graphene has a very low heat capacity, which means that it is possible to detect very small changes in its energy quickly. It is this speed in detecting the energy differences that makes it perfect for applications in measuring qubits.

Recently, researchers have produced a bolometer that can make measurements in well below a microsecond, as fast as the technology currently used to measure qubits. After these initial results, there is still a lot of optimisation that can be done to make the device even better.

Superconductivity at room temperature. For the first time, physicists at Rochester University have achieved superconductivity at room temperature—a major new milestone in the quest for superconductivity. Combining carbon, sulphur, and hydrogen to create microscopic samples of carbonaceous sulphur hydride, they found that the sample was superconducting at 15°C, although at a high pressure of 270 gigapascals. Obviously, it is still unuseable in everyday circumstances. The next step in the research will be to try to reduce the high pressure needed by tuning the chemical composition of the sample. The researchers believe a room-temperature, ambient-pressure superconductor will finally be within our grasp.

Lower noise level. Researchers at UNSW Sydney have demonstrated the lowest noise level on record for a semiconductor quantum bit. The team's research revealed that the presence of defects either within the silicon chip or at the interface to the surface were significant contributors to the charge noise. This was a surprise, as a lot of time was spent on optimising the quality of the silicon chip, but this showed that even a few impurities nearby can affect the noise.

By reducing the impurities in the silicon chip and positioning the atoms away from the surface and interfaces where most of the noise originates, the team was able to produce the record-breaking result. Results continue to show that silicon is a terrific material to host qubits. With the ability to engineer every aspect of the qubit environment, engineers are systematically proving that atom qubits in silicon are reproducible, fast, and stable. The next challenge is to move to isotopically pure crystalline Si-28 to capitalise on the long coherence times already demonstrated in this system.

Putting it all together

In principle, there are lots of ways to construct qubits. Collaboration between academia and industry is going to play a major role in how this shapes out. Depending on each of their strengths, academic labs can focus on supplying the core expertise necessary to get a quantum computer going while the industry can provide the engineering muscle to build such an intricate machine. Both are important parts.

To prepare for these developments, chip designers, chip-fabrication-process engineers, cryogenic-control specialists, experts in mass data handling, quantum-algorithm developers, and others will need to work together closely. By combining the efforts of academic institutions, research institutes, and commercial companies, we can succeed in building practical quantum computers, unleashing immense computing power for the future.

Quite a few world players have now come out with their models and have offered them for use by the interested parties. India has recently announced eighty billion rupees to the Department of Science and Technology (DST) with the aim of building the country's own 50-qubit quantum computer in the next four to five years.

To start off such a stupendous task and successfully complete the project, establishment of an entirely independent autonomous National Quantum Computing Academy (NQCA) under DST, with active industry participation for building our own quantum computer, is a prerequisite urgently needed today. NQCA will also guide and oversee all related activities including establishing hands-on learning workshops, school and collegiate education, and finally at post-graduate and doctoral level. **EFY**

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